EXPERIMENT – 11

Query with Joins – EquiJoin, InnerJoin, OuterJoin

INNER JOIN:

```
mysql>
mysql> -- EXPERIMENT 11: Joins - Equi, Inner, Outer
mysql> SELECT s.name, d.dept_name FROM Student s, Department d WHERE s.dept_id = d.dept_id;
+--------+------------------+
| name   | dept_name        |
+--------+------------------+
| Arun   | Computer Science |
| Bala   | Computer Science |
| Charu  | Electronics      |
| Divya  | Electronics      |
| Esther | Mechanical       |
| Farook | Mechanical       |
| Gowri  | Civil            |
| Harish | Civil            |
+--------+------------------+
8 rows in set (0.00 sec)

mysql>
mysql> SELECT s.name, c.course_name, e.grade
    -> FROM Student s
    -> INNER JOIN Enrollment e ON s.roll_no = e.roll_no
    -> INNER JOIN Course c ON e.course_id = c.course_id;
+--------+---------------------------+-------+
| name   | course_name               | grade |
+--------+---------------------------+-------+
| Arun   | Advanced Database Systems | A     |
| Bala   | Advanced Database Systems | B     |
| Arun   | Operating Systems         | B     |
| Charu  | Digital Electronics       | A     |
| Divya  | Microprocessors           | C     |
| Esther | Thermodynamics            | B     |
| Farook | Thermodynamics            | C     |
| Gowri  | Structural Analysis       | A     |
| Harish | Structural Analysis       | B     |
+--------+---------------------------+-------+
9 rows in set (0.00 sec)
```

EQUIJOIN & OUTERJOIN:

```
mysql>
mysql> SELECT d.dept_name, s.name FROM Department d LEFT JOIN Student s ON d.dept_id = s.dept_id;
+------------------+--------+
| dept_name        | name   |
+------------------+--------+
| Aerospace        | NULL   |
| Civil            | Gowri  |
| Civil            | Harish |
| Computer Science | Arun   |
| Computer Science | Bala   |
| Electronics      | Charu  |
| Electronics      | Divya  |
| Mechanical       | Esther |
| Mechanical       | Farook |
+------------------+--------+
9 rows in set (0.00 sec)

mysql> SELECT d.dept_name, s.name FROM Department d RIGHT JOIN Student s ON d.dept_id = s.dept_id;
+------------------+--------+
| dept_name        | name   |
+------------------+--------+
| Computer Science | Arun   |
| Computer Science | Bala   |
| Electronics      | Charu  |
| Electronics      | Divya  |
| Mechanical       | Esther |
| Mechanical       | Farook |
| Civil            | Gowri  |
| Civil            | Harish |
+------------------+--------+
8 rows in set (0.00 sec)
```